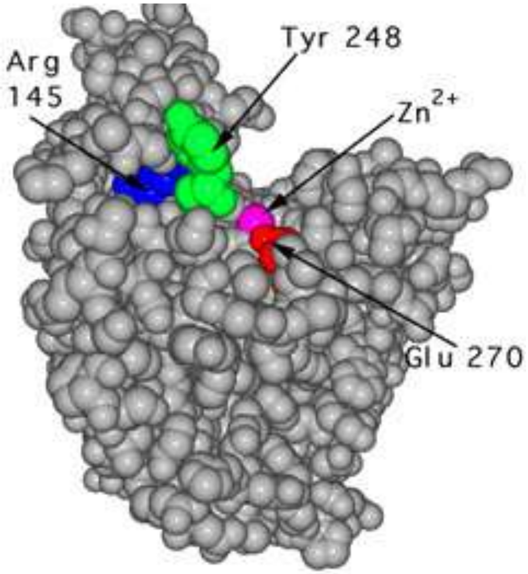
Enzymology- An overview



Biochemistry for Medics
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Enzymes- Introduction

- Biologic polymers that catalyze the chemical reactions
- Enzymes are neither consumed nor permanently altered as a consequence of their participation in a reaction
- With the exception of catalytic RNA molecules, or ribozymes, enzymes are proteins
- In addition to being highly efficient, enzymes are also extremely selective catalysts

Classification of Enzymes

Class	Reactions catalyzed
Oxidoreductases	oxidation-reduction
Transferases	transfer of moieties such as glycosyl, methyl, or phosphoryl groups
Hydrolases	catalyze hydrolytic cleavage
Lyases	add/remove atoms to/from a double bond
Isomerases	geometric or structural changes within a molecule
Ligases	joining together of two molecules coupled to the hydrolysis of ATP

Cofactors

- Cofactors can be subdivided into two groups: metals and small organic molecules
- Cofactors that are small organic molecules are called coenzymes.
- Most common cofactor are also metal ions
- If tightly bound, the cofactors are called prosthetic groups
- Loosely bound Cofactors serve functions similar to those of prosthetic groups but bind in a transient, dissociable manner either to the enzyme or to a substrate

Prosthetic group

- Tightly integrated into the enzyme structure by covalent or non-covalent forces. e.g;
  - Pyridoxal phosphate
  - Flavin mononucleotide(FMN)
  - Flavin adenine dinucleotide(FAD)
  - Thiamin pyrophosphate (TPP)
  - Biotin
  - Metal ions - Co, Cu, Mg, Mn, Zn
- Metals are the most common prosthetic groups

Role of metal ions

- Enzymes that contain tightly bound metal ions are termed – Metalloenzymes

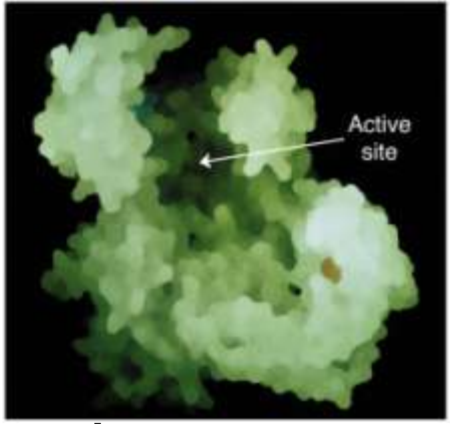
- Enzymes that require metal ions as loosely bound cofactors are termed as metal-activated enzymes

- Metal ions facilitate
  - Binding and orientation of the substrate
  - Formation of covalent bonds with reaction intermediates
  - Interact with substrate to render them more electrophilic or nucleophilic

Coenzymes

- **Coenzymes** serve as recyclable shuttles—or group transfer agents—that transport many substrates from their point of generation to their point of utilization.
- The water-soluble B vitamins supply important components of numerous coenzymes
- chemical moieties transported by coenzymes include hydrogen atoms or hydride ions, methyl groups (folates), acyl groups (coenzyme A), and oligosaccharides (dolichol).

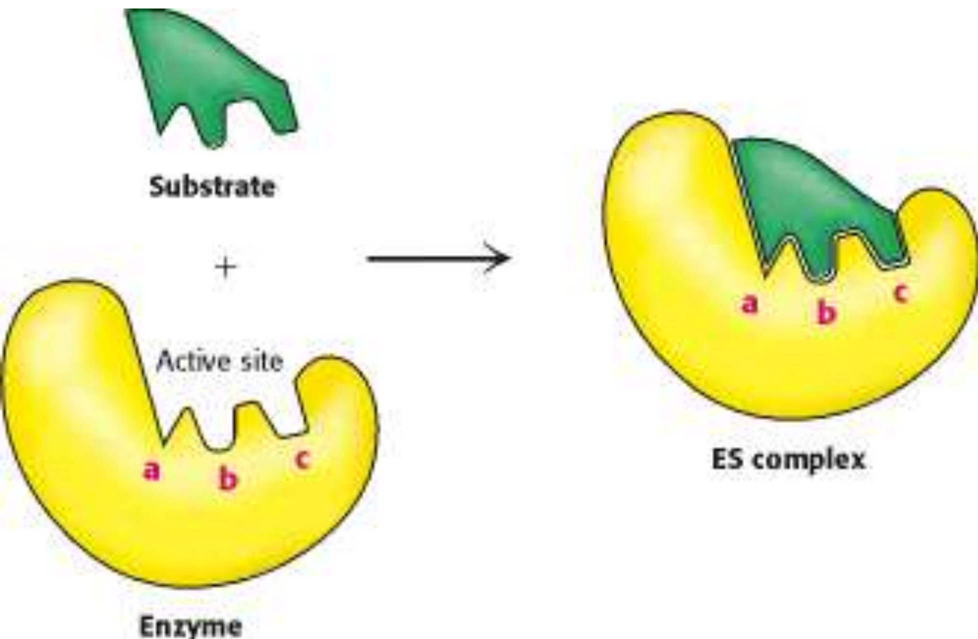
Active Site



- Takes the form of a cleft or pocket
- Takes up a relatively small part of the total volume of an enzyme
- Substrates are bound to enzymes by multiple weak attractions
- The specificity of binding depends on the precisely defined arrangement of atoms in an active site
- The active sites of multimeric enzymes are located at the interface between subunits and recruit residues from more than one monomer

Enzyme substrate binding

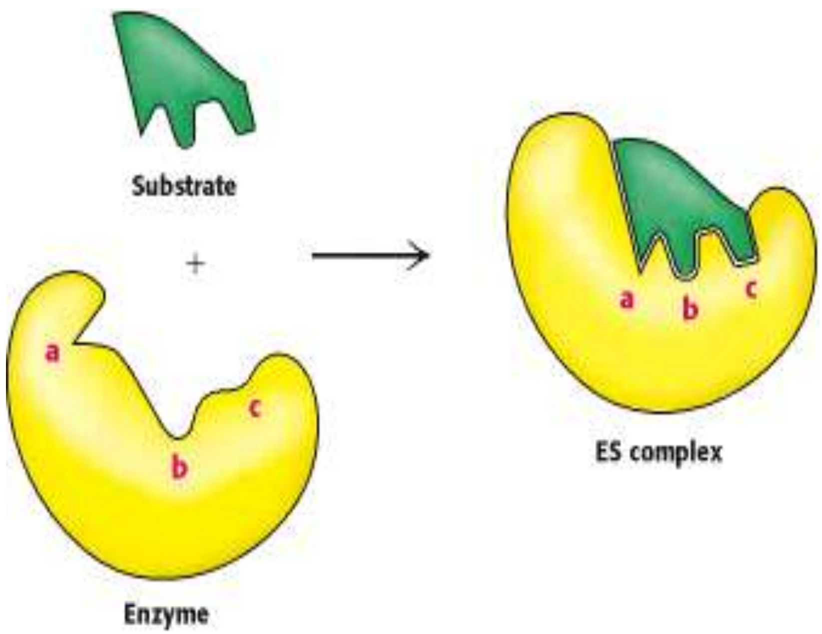
- **Two models have been proposed to explain how an enzyme binds its substrate: the lock-and -key model and the induced-fit model.**



Lock-and-Key Model of Enzyme-Substrate Binding In this model, the active site of the unbound enzyme is complementary in shape to the substrate.

"lock and key model" accounted for the exquisite specificity of enzyme-substrate interactions, the implied rigidity of the enzyme's active site failed to account for the dynamic changes that accompany catalysis.

Enzyme substrate binding



Induced-Fit Model of Enzyme-Substrate Binding

In this model, the enzyme changes shape on substrate binding. The active site forms a shape complementary to the substrate only after the substrate has been bound. When a substrate approaches and binds to an enzyme they induce a conformational change, a change analogous to placing a hand (substrate) into a glove (enzyme).

Mechanism of Action of Enzymes

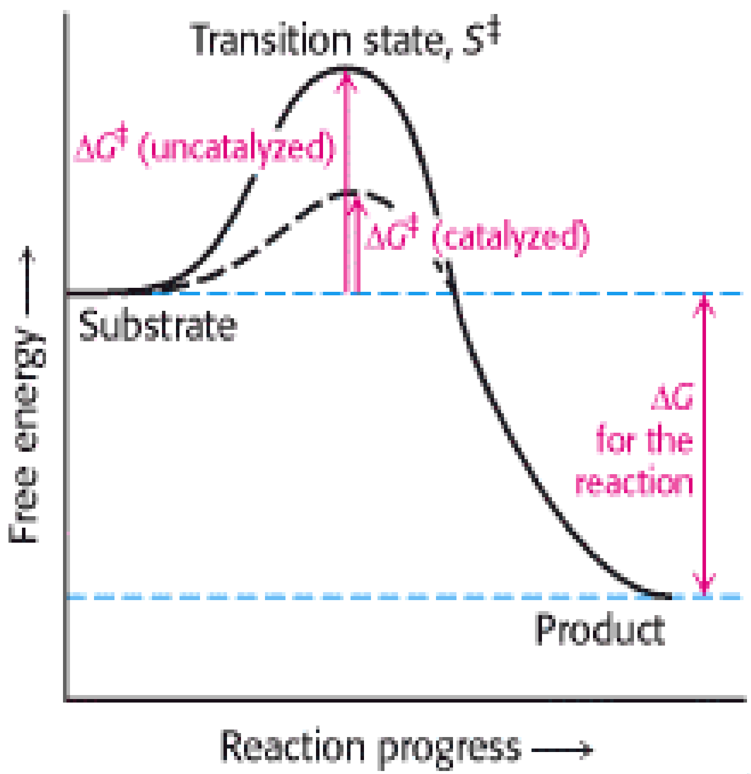
- Enzymes are catalysts and increase the speed of a chemical reaction without themselves undergoing any permanent chemical change. They are neither used up in the reaction nor do they appear as reaction products.
- The basic enzymatic reaction can be represented as follows

$$S \quad + \quad E \longrightarrow P \quad + \quad E$$

- where E represents the enzyme catalyzing the reaction, S the substrate, the substance being changed, and P the product of the reaction.
-
- **The mechanism of action of enzymes can be explained by two perspectives-**
- **Thermodynamic changes**
- **Processes at the active site**

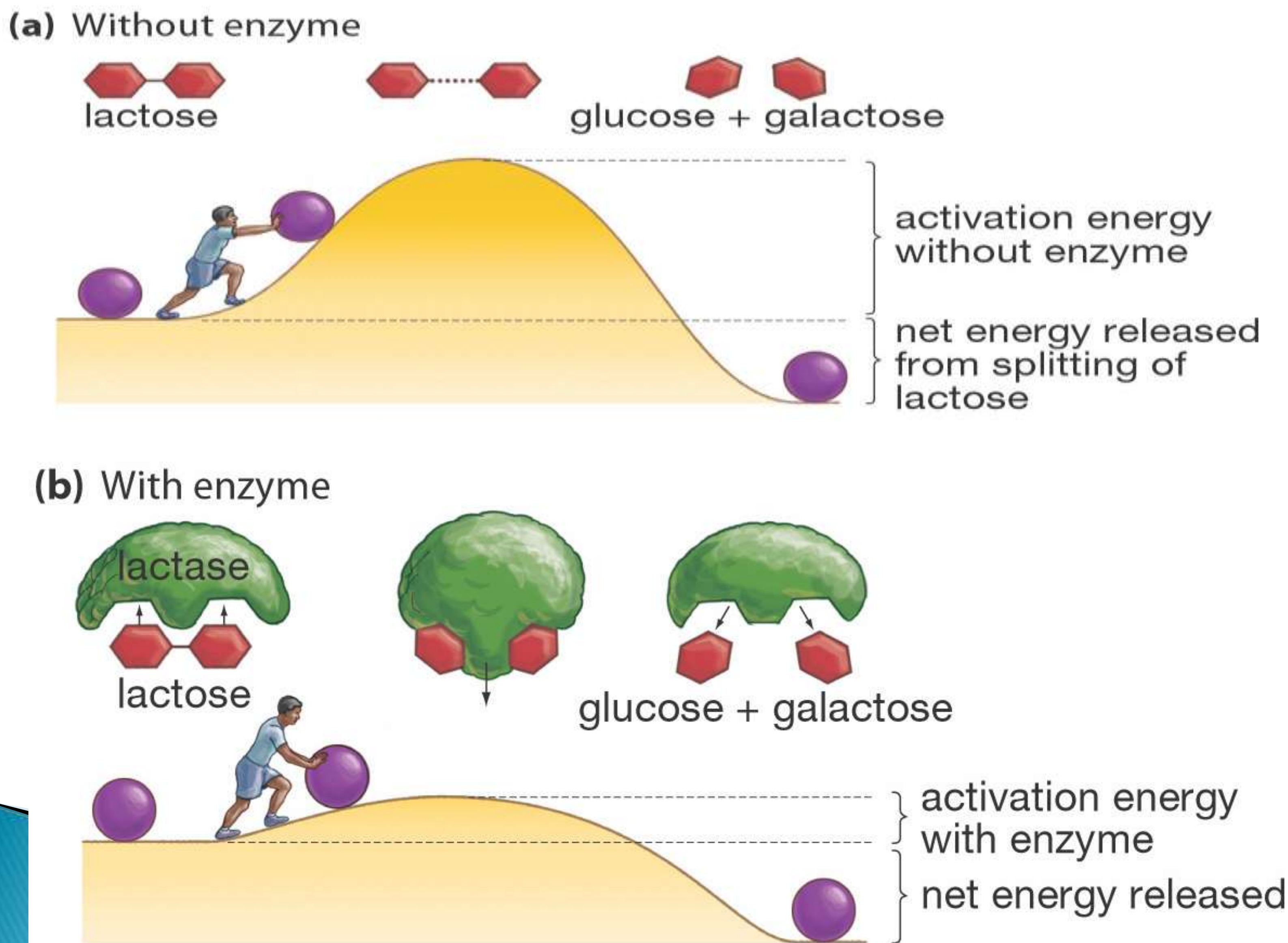
1) Thermodynamic changes

- All enzymes accelerate reaction rates by providing transition states with a lowered ΔG F for formation of the transition states.



The lower activation energy means that more molecules have the required energy to reach the transition state.

Thermodynamic changes–overview



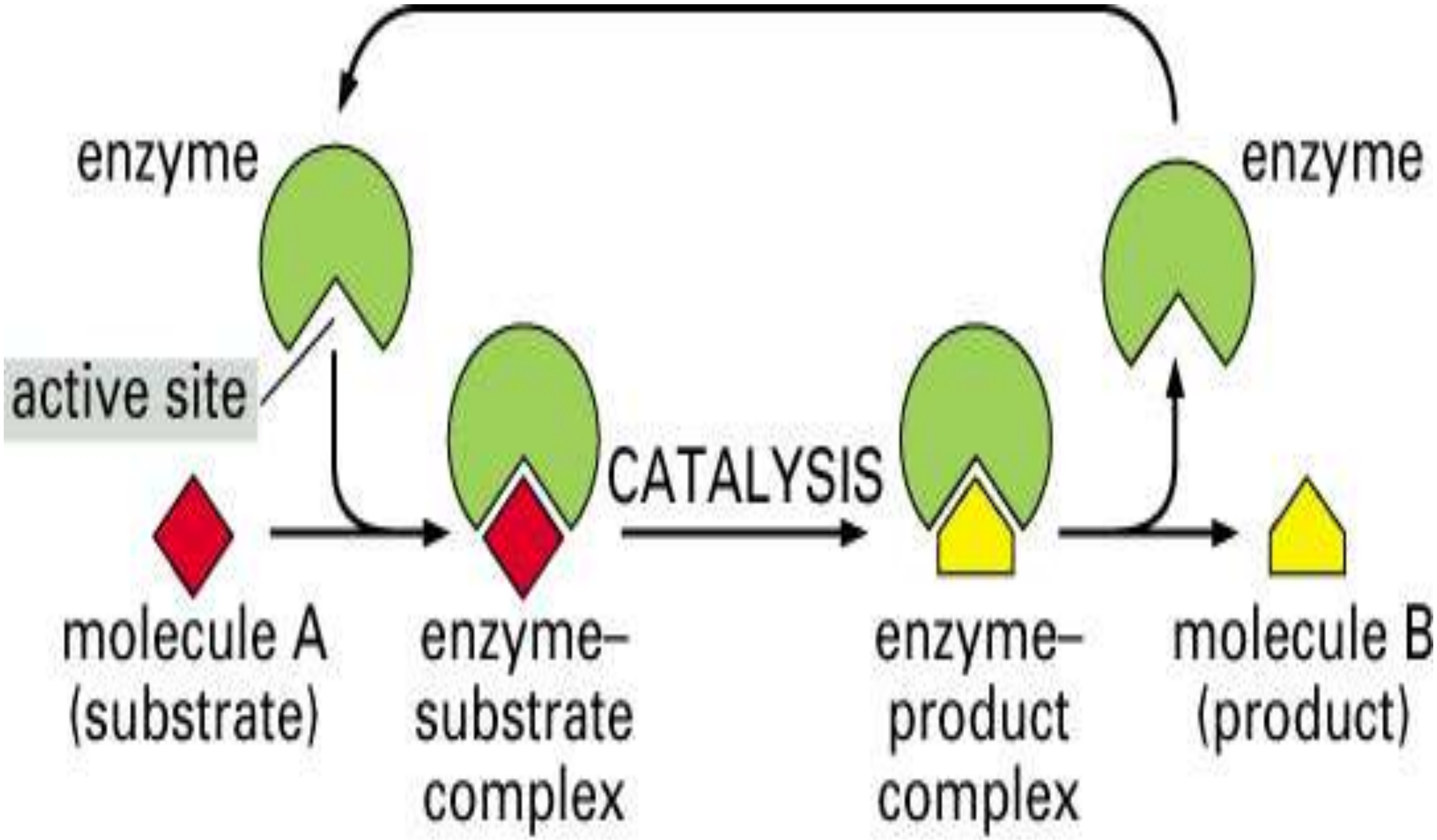
2) Processes at the active site

1. Catalysis by proximity – **for the molecules to react they must come within bond-forming distance of one another.** When an enzyme binds substrate molecules at its active site, it creates a region of high local substrate concentration. Enzyme-substrate interactions orient reactive groups and bring them into proximity with one another.
2. Acid base catalysis – **the ionizable functional groups of aminoacyl side chains of prosthetic groups contribute to catalysis by acting as acids or bases**

2) Processes at the active site

- Catalysis by strain – enzymes that catalyze the lytic reactions involve breaking a covalent bond typically bind their substrates in a configuration slightly unfavorable for the bond that will undergo cleavage
- Covalent Catalysis – involves the formation of a covalent bond between the enzyme and one or more substrates which introduces a new reaction pathway whose activation energy is lower

Enzyme Catalysis– overview



Enzyme Specificity

- In general, there are four distinct types of specificity:
- **Absolute specificity** – the enzyme will catalyze only one reaction.
- **Group specificity** – the enzyme will act only on molecules that have specific functional groups, such as amino, phosphate and methyl groups
- **Linkage specificity** – the enzyme will act on a particular type of chemical bond regardless of the rest of the molecular structure
- **Stereo chemical specificity** – the enzyme will act on a particular steric or optical isomer.

Enzyme Kinetics

- **Enzyme kinetics** is the field of biochemistry concerned with the quantitative measurement of the rates of enzyme-catalyzed reactions and the systematic study of factors that affect these rates.
- The rate of a chemical reaction is described by the number of molecules of reactant(s) that are converted into product(s) in a specified time period.
- Reaction rate is always dependent on the concentration of the chemicals involved in the process and on rate constants that are characteristic of the reaction.

Enzyme activity

- The catalytic action of an enzyme, its activity, is measured by determining the increase in the reaction rate under precisely defined conditions—i. e., the difference between the turnover of the catalyzed reaction and uncatalyzed reaction in a specific time interval

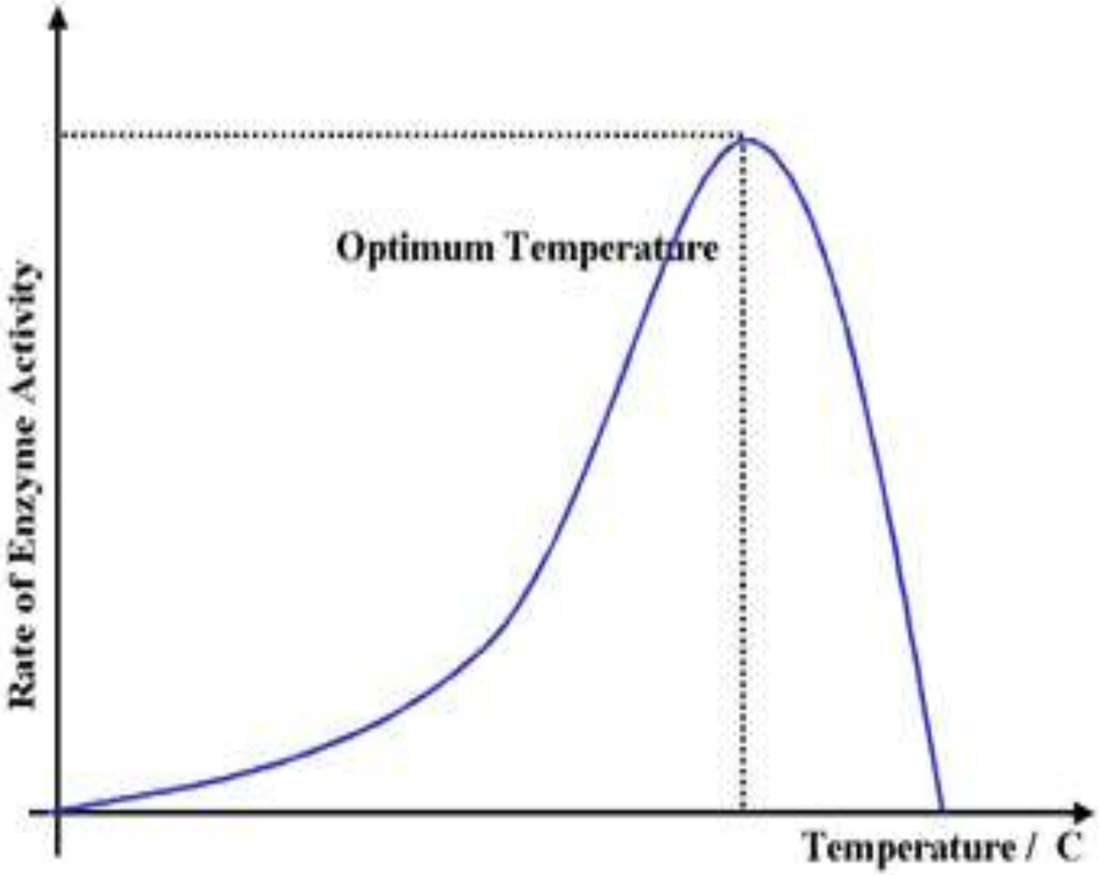
- Normally, reaction rates are expressed as the *change in concentration per unit of time* (mol 1^{-1} s^{-1}).

- The catalytic activity of an enzyme is
  independent of the volume, the unit used for enzymes is usually *turnover per unit time*, expressed in –
  katal (kat, mol s^{-1}).
  However, the international unit U is still more commonly used (μmol turnover min^{-1};
  1 U = 16.7 nkat).

Factors affecting Enzyme activity

- Numerous factors affect the reaction rate-
- **Temperature**
- The reaction rate increases with temperature to a maximum level, then abruptly declines with further increase of temperature
- Most animal enzymes rapidly become denatured at temperatures above 40^{o}C
- The optimal temperatures of the enzymes in higher organisms rarely exceed 50 °C
- The **Q_{10}, or temperature coefficient,** is the factor by which the rate of a biologic process increases for a 10 °C increase in temperature.

Effect of Temperature

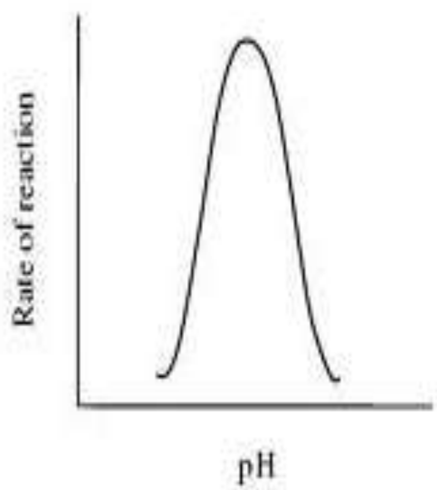


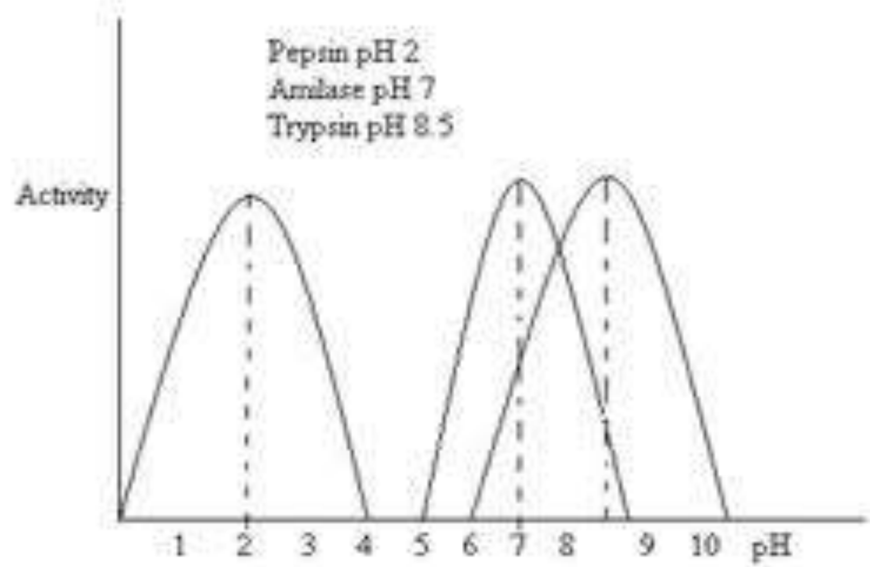
For mammals and other homoeothermic organisms, changes in enzyme reaction rates with temperature assume physiologic importance only in circumstances such as fever or hypothermia.

Effect of pH on enzyme activity

- The rate of almost all enzyme-catalyzed reactions exhibits a significant dependence on hydrogen ion concentration
- Most intracellular enzymes exhibit optimal activity at pH values between 5 and 9.
- The relationship of activity to hydrogen ion concentration reflects the balance between enzyme denaturation at high or low pH and effects on the charged state of the enzyme, the substrates, or both.

Effect of pH on enzyme activity

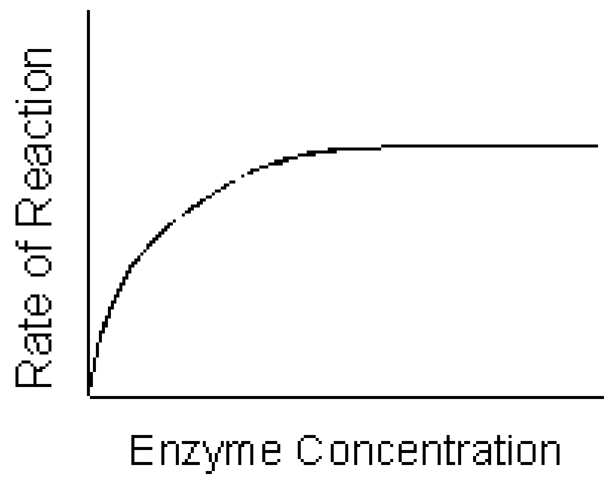




Except for Pepsin, acid phosphatase and alkaline phosphatase, most enzyme have optimum pH between 5 to 9

Effect of enzyme concentration

As the amount of enzyme is increased, the rate of reaction increases. If there are more enzyme molecules than are needed, adding additional enzyme will not increase the rate. Reaction rate therefore increases as enzyme concentration increases but then it levels off.

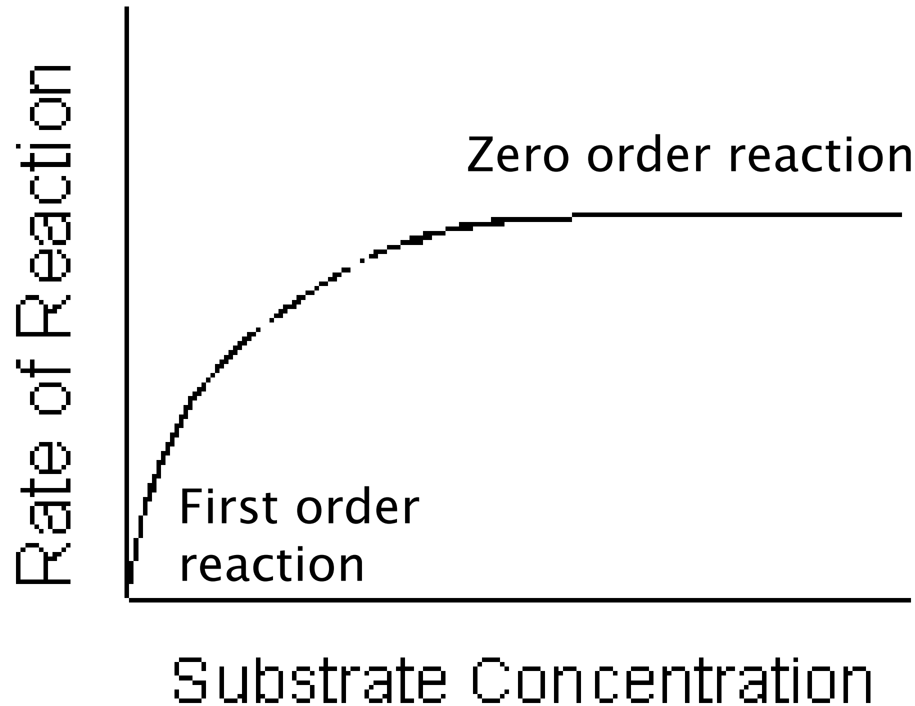


Effect of substrate concentration

- At lower concentrations, the active sites on most of the enzyme molecules are not filled because there is not much substrate. Higher concentrations cause more collisions between the molecules. The rate of reaction increases(First order reaction).
- The maximum velocity of a reaction is reached when the active sites are almost continuously filled. Increased substrate concentration after this point will not increase the rate. Reaction rate therefore increases as substrate concentration is increased but it levels off (Zero order reaction)

Effect of substrate concentration

The shape of the curve that relates activity to substrate concentration is hyperbolic.



Michaelis-Menten Kinetics

- The Michaelis-Menten equation is a quantitative description of the relationship among the rate of an enzyme- catalyzed reaction [v_1], the concentration of substrate [S] and two constants, V max and km (which are set by the particular equation).
- The symbols used in the Michaelis-Menten equation refer to the reaction rate [v_1], maximum reaction rate (V max), substrate concentration [S] and the Michaelis-Menten constant (km).

Michaelis–Menten equation

$$v_1 = \frac{V_{max}[S]}{\{K_m + [S]\}}$$

The dependence of initial reaction velocity on [S] and Km may be illustrated by evaluating the Michaelis–Menten equation under three conditions.
(1) When [S] is much less than km, the term km + [S] is essentially equal to km. Since V max and km are both constants, their ratio is a constant (k). In other words, when [S] is considerably below km, V max is proportionate to k[S]. The initial reaction velocity therefore is directly proportionate to [S].

$$v_1 = \frac{V_{max}[S]}{K_m + [S]} \quad v_1 \approx \frac{V_{max}[S]}{K_m} \approx \left(\frac{V_{max}}{K_m}\right)[S]$$

Michaelis-Menten equation

- **(2)** When [S] is much greater than km, the term km + [S] is essentially equal to [S]. Replacing km + [S] with [S] reduces equation to

$$v_i = \frac{V_{max}[S]}{K_m + [S]} \quad v_i \approx \frac{V_{max}[S]}{[S]} \approx V_{max}$$

- Thus, when [S] greatly exceeds km, the reaction velocity is maximal (V max) and unaffected by further increases in substrate concentration.

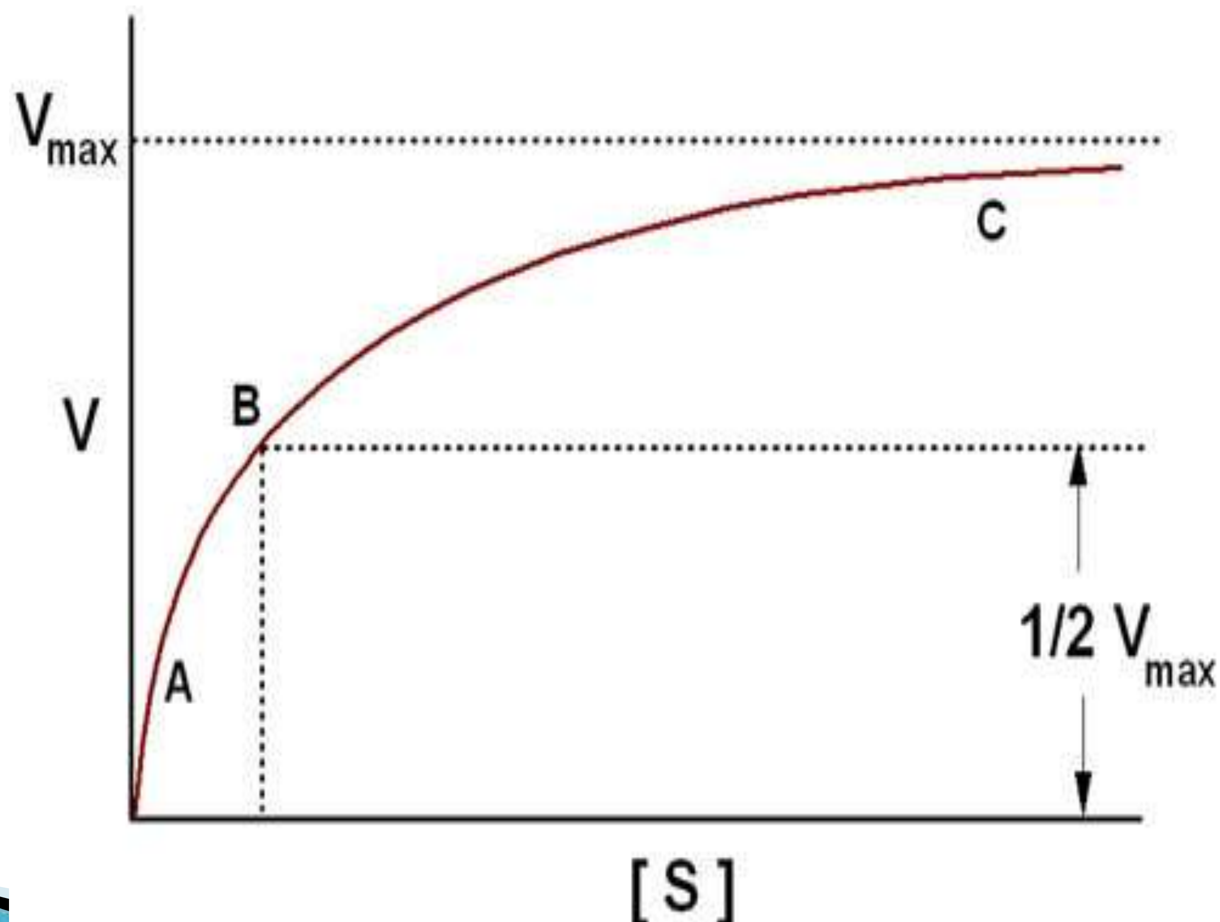
Michaelis-Menten equation

- (3) When [S] = km

$$v_i = \frac{V_{max}[S]}{K_m + [S]} = \frac{V_{max}[S]}{2[S]} = \frac{V_{max}}{2}$$

- Equation states that when [S] equals km, the initial velocity is half-maximal. Equation also reveals that km is a constant and may be determined experimentally from—the substrate concentration at which the initial velocity is half-maximal.

Plot of substrate concentration versus reaction velocity



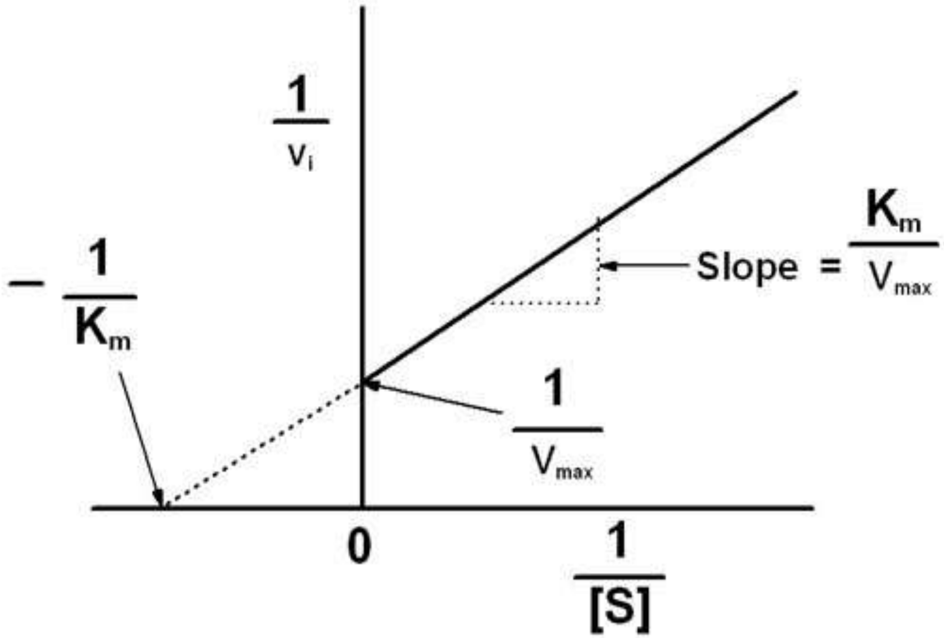
Lineweaver-Burk Plot

- **A Linear Form of the Michaelis-Menten Equation Is Used to determine km & V max**

$$v_i = \frac{V_{max}[S]}{K_m + [S]}$$ Invert $$\frac{1}{v_i} = \frac{K_m + [S]}{V_{max}[S]}$$ factor $$\frac{1}{v_i} = \frac{K_m}{V_{max}[S]} + \frac{[S]}{V_{max}[S]}$$ and simplify

$$\frac{1}{v_i} = \left(\frac{K_m}{V_{max}}\right)\frac{1}{[S]} + \frac{1}{V_{max}}$$

Lineweaver–Burk Plot



- A plot of $1/v_i$ as *y* as a function of $1/[S]$ as *x* therefore gives a straight line whose *y* intercept is 1/ **V max** and whose slope is **km / V max.** Such a plot is called a **double reciprocal** or **Lineweaver–Burk plot**

Km and its sinificance

- The Michaelis constant K_m is the substrate concentration at which v_i is half the maximal velocity ($V_{max}/2$) attainable at a particular concentration of enzyme
- It is specific and constant for a given enzyme under defined conditions of time , temperature and p H
- K_m determines the affinity of an enzyme for its substrate, lesser the Km for is the affinity and vice versa
- K_m value helps in determining the true substrate for the enzyme.

Enzyme Inhibition

- Inhibitors are chemicals that reduce the rate of enzymic reactions
- The are usually specific and they work at low concentrations
- They block the enzyme but they do not usually destroy it
- Many drugs and poisons are inhibitors of enzymes in the nervous system
- Inhibitors of the catalytic activities of enzymes provide both pharmacologic agents and research tools for study of the mechanism of enzyme action.

The effect of enzyme inhibition

- **Irreversible inhibitors:** Combine with the functional groups of the amino acids in the active site, irreversibly
- **Reversible inhibitors:** These can be washed out of the solution of enzyme by dialysis.

Applications of inhibitors

- **Negative feedback**: end point or end product inhibition
- **Poisons** snake bite, plant alkaloids and nerve gases
- **Medicine** antibiotics, sulphonamides, sedatives and stimulants

Enzyme Inhibition

Classification

- Inhibitors can be classified based upon their site of action on the enzyme,
- on whether they chemically modify the enzyme,
- or on the kinetic parameters they influence.

Types of Enzyme inhibition

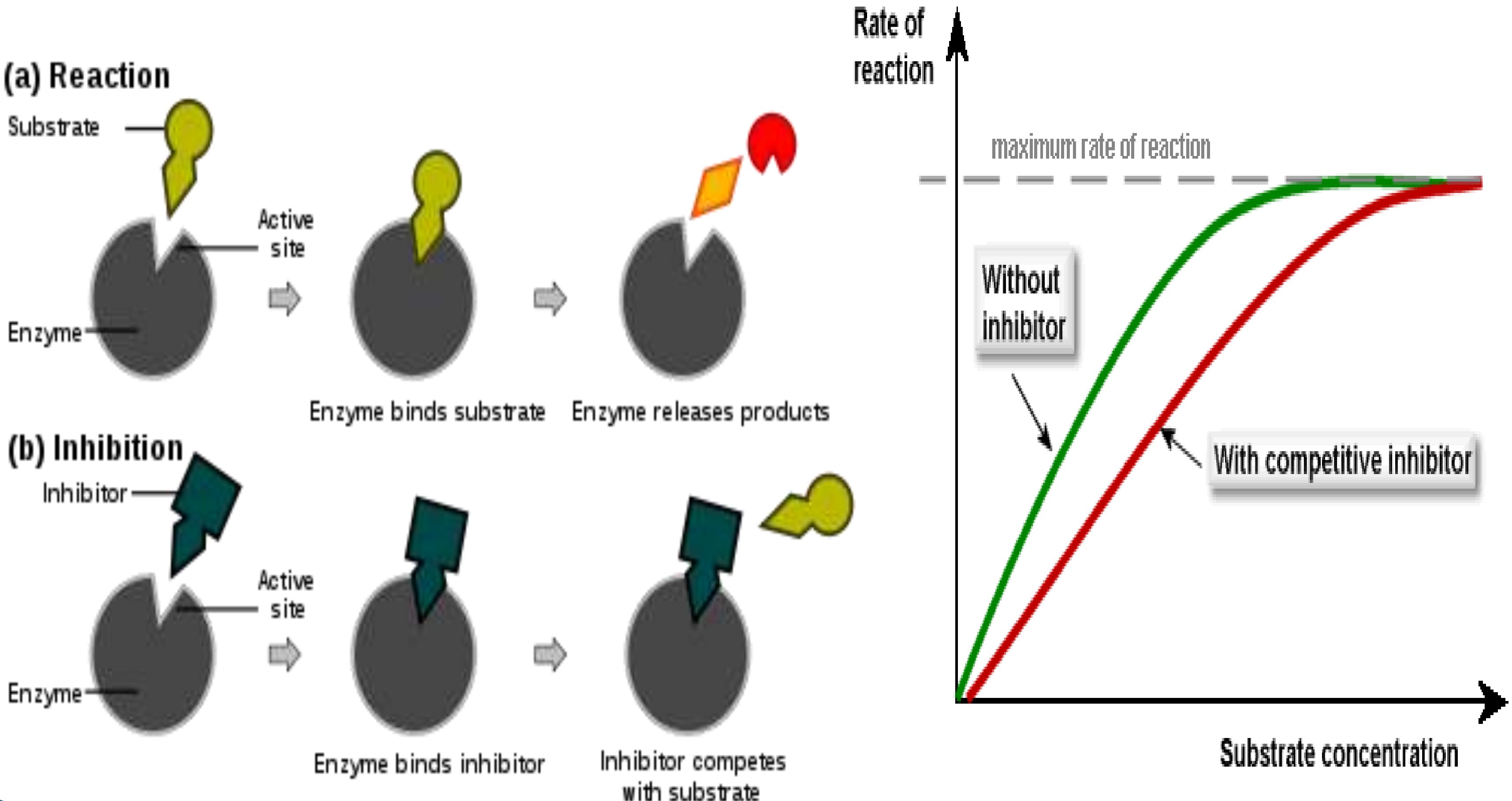
- Competitive Enzyme Inhibition
- Non Competitive Enzyme Inhibition
- Uncompetitive Enzyme Inhibition
- Suicide Enzyme Inhibition
- Allosteric Enzyme Inhibition

Competitive Enzyme Inhibition

A competitive inhibitor

- Has a structure similar to substrate(Structural Analog)
- Occupies active site
- Competes with substrate for active site
- Has effect reversed by increasing substrate concentration
- Vmax remains same but Km is increased

Competitive Enzyme Inhibition



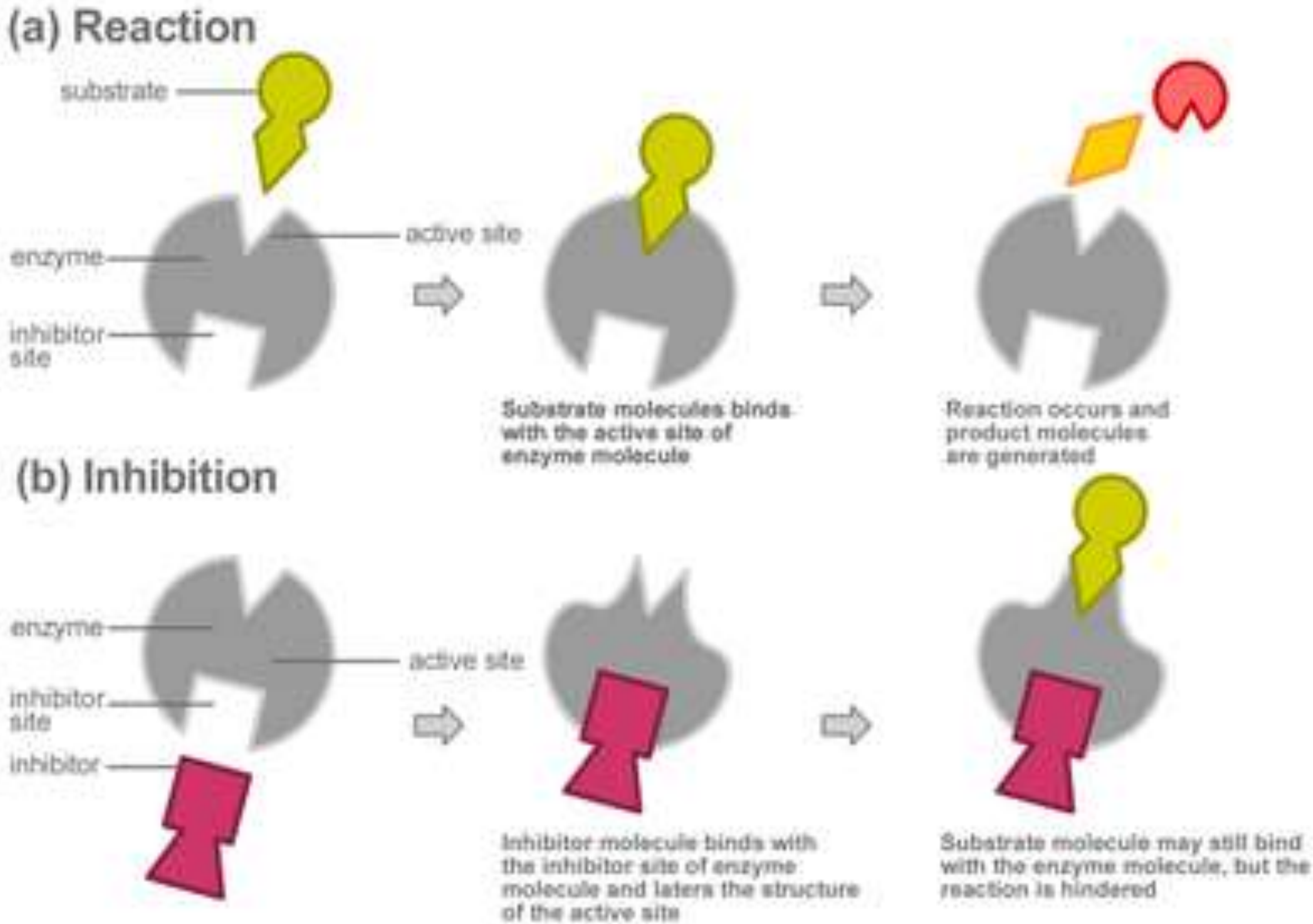
Clinical significance of competitive enzyme Inhibitors

Drug	Enzyme Inhibited	Clinical Use
Dicoumarol	Vitamin K Epoxide Reductase	Anticoagulant
Sulphonamide	Pteroid Synthetase	Antibiotic
Trimethoprim	Dihydrofolate reductase	Antibiotic
Pyrimethamine	Dihydrofolate reductase	Antimalarial
Methotrexate	Dihydrofolate reductase	Anticancer
Lovastatin	HMG Co A Reductase	Cholesterol Lowering drug
Alpha Methyl Dopa	Dopa decarboxylase	Antihypertensive
Neostigmine	Acetyl Cholinesterase	Myasthenia Gravis

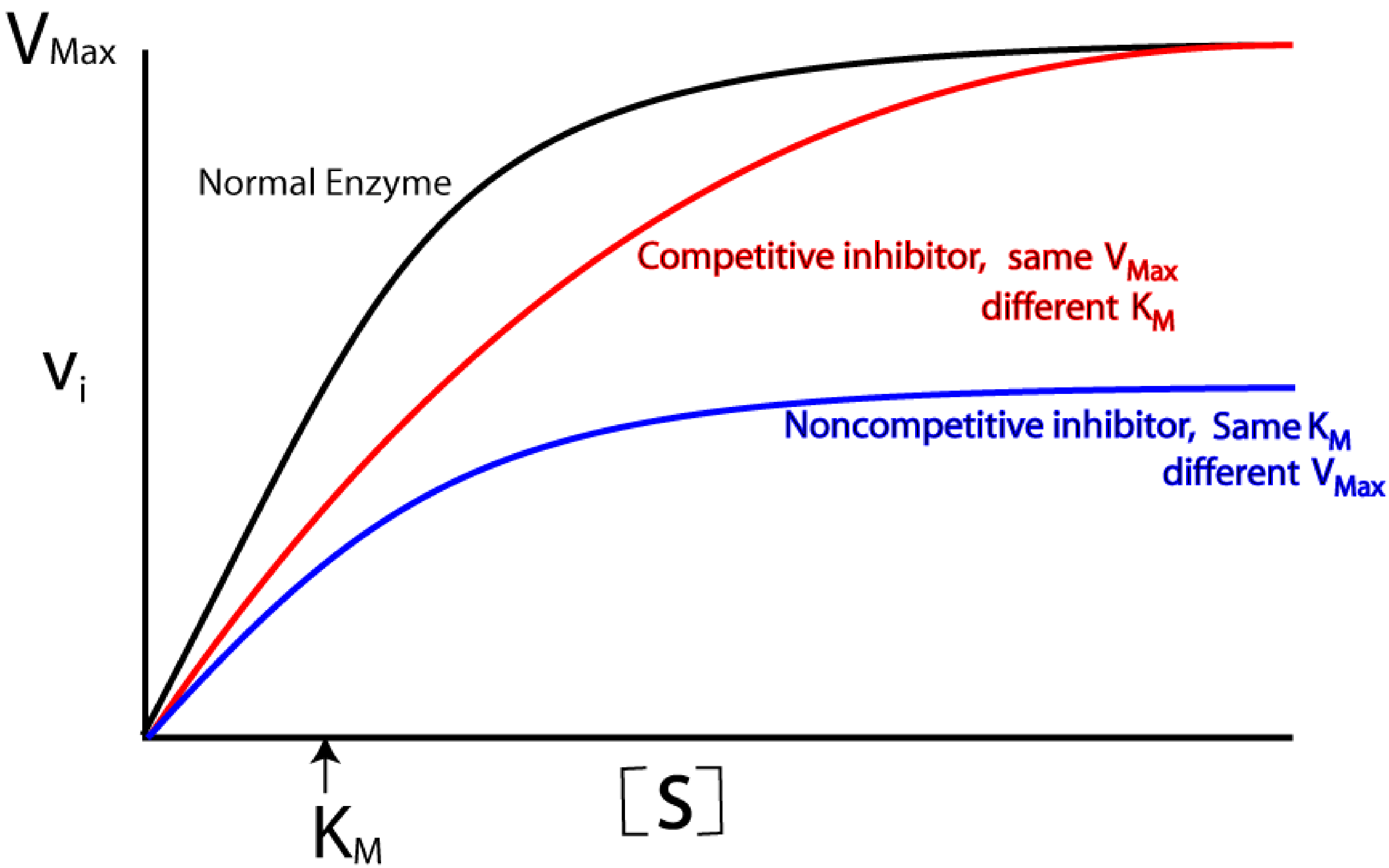
Non Competitive Enzyme Inhibition

- Noncompetitive inhibitors bind enzymes at sites distinct from the substrate-binding site
- Generally bear little or no structural resemblance to the substrate
- Binding of the inhibitor does not affect binding of substrate
- Formation of both EI and EIS complexes is therefore possible
- The enzyme-inhibitor complex can still bind substrate, its efficiency at transforming substrate to product, reflected by V_{max}, is decreased.

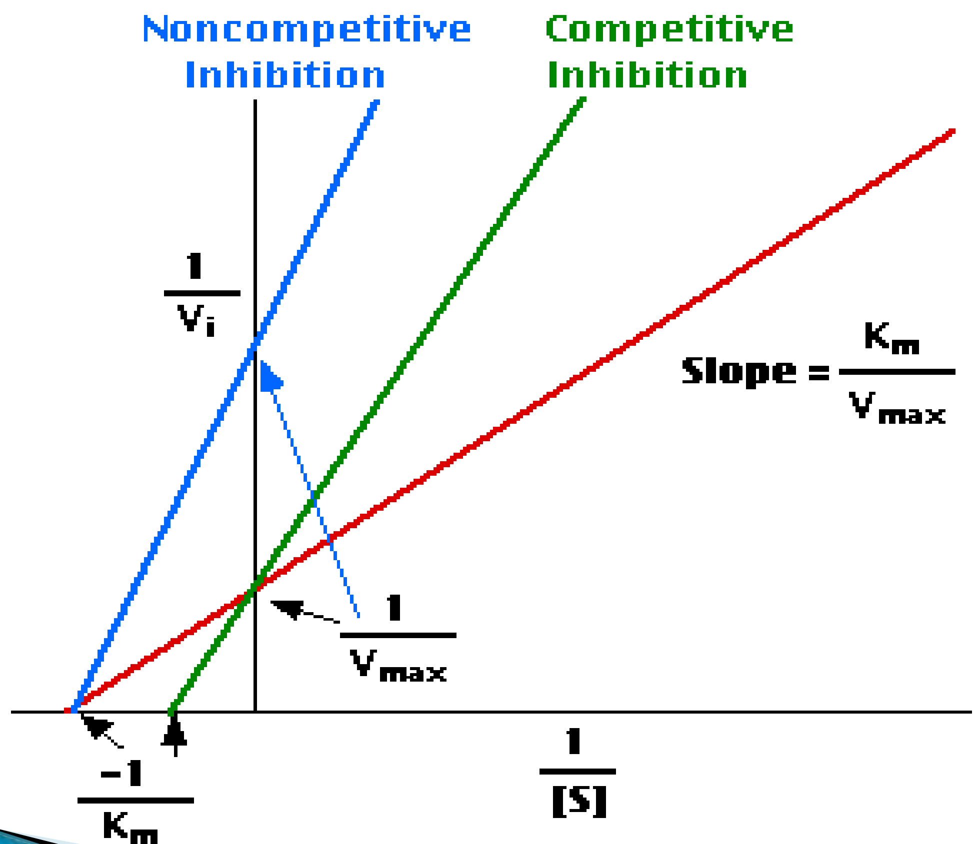
Non Competitive Enzyme Inhibition



Non Competitive Enzyme Inhibition



Lineweaver Burk plot



In the presence of a competitive inhibitor, V_{max} can still be reached if sufficient substrate is available, one-half V_{max} requires a higher **[S]** than before and thus K_m is larger.

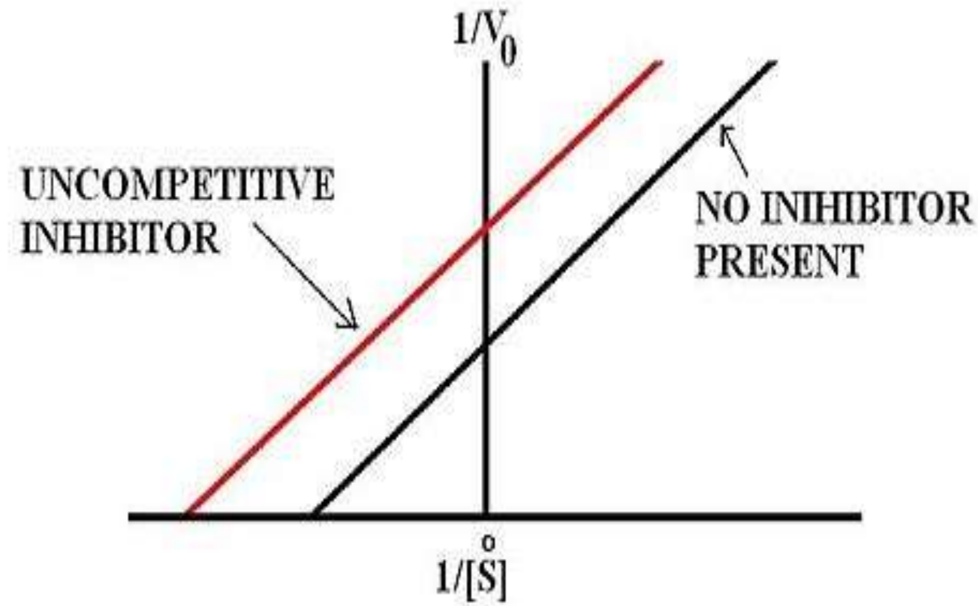
With noncompetitive inhibition, enzyme rate (velocity) is reduced for all values of **[S]**, including V_{max} and one-half V_{max} but K_m remains unchanged

Examples of Non Competitive Enzyme Inhibitors

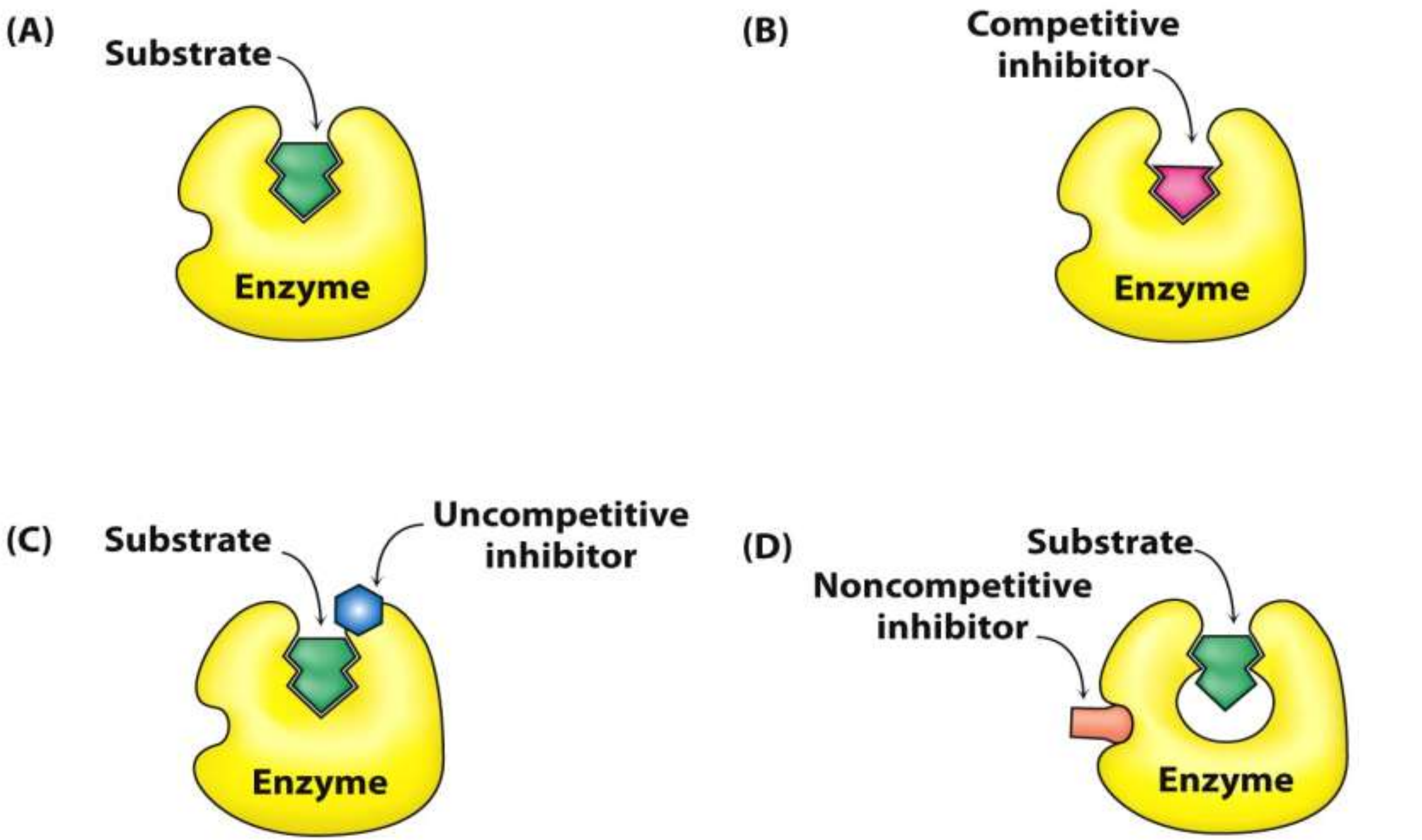
- Cyanide inhibits cytochrome oxidase
- Fluoride inhibits Enolase and hence glycolysis
- Iodoacetate inhibits enzymes having SH groups in their active sites
- BAL (British Anti Lewisite, dimercaprol) is used as an antidote for heavy metal poisoning
  - Heavy metals act as enzyme poisons by reacting with the SH groups
  - BAL has several SH groups with which the heavy metal ions bind and thereby their poisonous effects are reduced

Uncompetitive Enzyme Inhibition

- Inhibitor binds to enzyme-substrate complex
- Both Vmax and Km are decreased
- e.g ; Inhibition of placental alkaline phosphatase (Regan isoenzyme) by phenylalanine



Competitive V/S Non competitive V/S Uncompetitive Enzyme Inhibition



Suicide Inhibition

- Irreversible inhibition
- Structural analog of the substrate is converted to more effective inhibitor with the help of enzyme to be inhibited
- The new product irreversibly binds to the enzyme and inhibits further reaction
- e.g;
  - Ornothine decarboxylase – is irreversibly inhibited by difluormethyl ornithine, as a result multiplication of parasite is arrested . Used against trypanosome in sleeping sickness

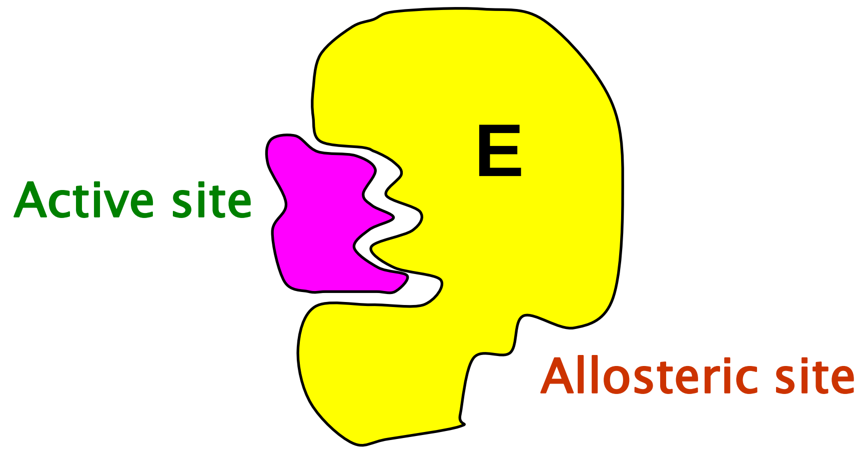
Suicide Inhibition

- Allopurinol is oxidized by Xanthine oxidase to alloxanthine which is a strong inhibitor of Xanthine Oxidase
- Aspirin action is based on suicide inhibition
  - Acetylates a serine residue in the active center of cyclo oxygenase . Thus PG synthesis is inhibited so inflammation subsides
- Disulfiram – used in treatment of alcoholism
  - Drug irreversibly inhibits the enzyme aldehyde dehydrogenase preventing further oxidation of acetaldehyde which produces sickening effects leading to aversion to alcohol.

Allosteric Inhibition

- Allosteric means **“other site”**

These enzymes have one catalytic site (Active site) where the substrate binds and another separate allosteric site where the modifier binds.
The allosteric sites may or may not be physically adjacent.
The binding of the modifier may enhance (Positive modifier) or inhibit (Negative modifier) the enzyme activity.



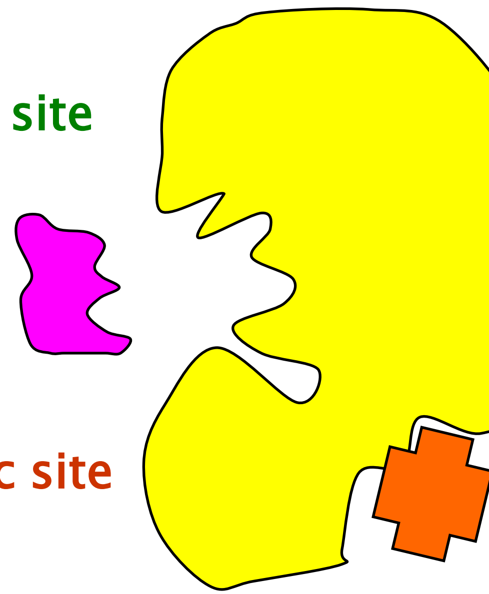
Allosteric Inhibition– Salient features

- Inhibitor is not a substrate analogue
- Partially reversible, when excess substrate is added
- Km is usually increased(K series enzymes)
- Vmax is reduced(V series enzymes)
- When the inhibitor binds the allosteric site, the configuration of the active site is changed so that the substrate can not bind properly.
- Most allosteric enzymes possess quaternary structure.

Switching off

Substrate
cannot fit into the active site

Inhibitor fits into allosteric site



Inhibitor molecule

When the inhibitor is present it fits into its site and there is a **conformational change** in the enzyme molecule
The enzyme’s molecular shape changes
The active site of the substrate changes
The substrate cannot bind with the substrate.
The reaction slows down

When the inhibitor concentration diminishes the enzyme’s conformation changes back to its active form. This is **not** competitive inhibition but it is reversible

Allosteric Modification (Positive)

- **Phosphofructokinase** This enzyme an **active site** for fructose-6-phosphate molecules to bind with another phosphate group
- It has an **allosteric site** for ATP molecules, the inhibitor
- When the cell consumes a lot of ATP the level of ATP in the cell falls
- No ATP binds to the **allosteric site** of phosphofructokinase
- The enzyme's conformation (shape) changes and the **active site** accepts substrate molecules

Allosteric Modification (Negative)

- **Phosphofructokinase**
- The respiration pathway accelerates and ATP (the final product) builds up in the cell
- As the ATP increases, more and more ATP fits into the **allosteric site** of the phosphofructokinase molecules
- The enzyme's conformation changes again and stops accepting substrate molecules in the **active site**
- Respiration slows down

Feed back(End point) Inhibition

Cell processes consist of series of pathways controlled by enzymes

- $A \xrightarrow{\mathbf{e_A}} B \xrightarrow{\mathbf{e_B}} C \xrightarrow{\mathbf{e_C}} D \xrightarrow{\mathbf{e_D}} E \xrightarrow{\mathbf{e_F}} F$

Each step is controlled by a different enzyme ($\mathbf{e_A}$, $\mathbf{e_B}$, $\mathbf{e_C}$ etc)

This is possible because of enzyme specificity

Feed back(End point) inhibition

- The first step (controlled by $\mathbf{e_A}$) is often controlled by the end product (**F**)
- Therefore **negative feedback** is possible

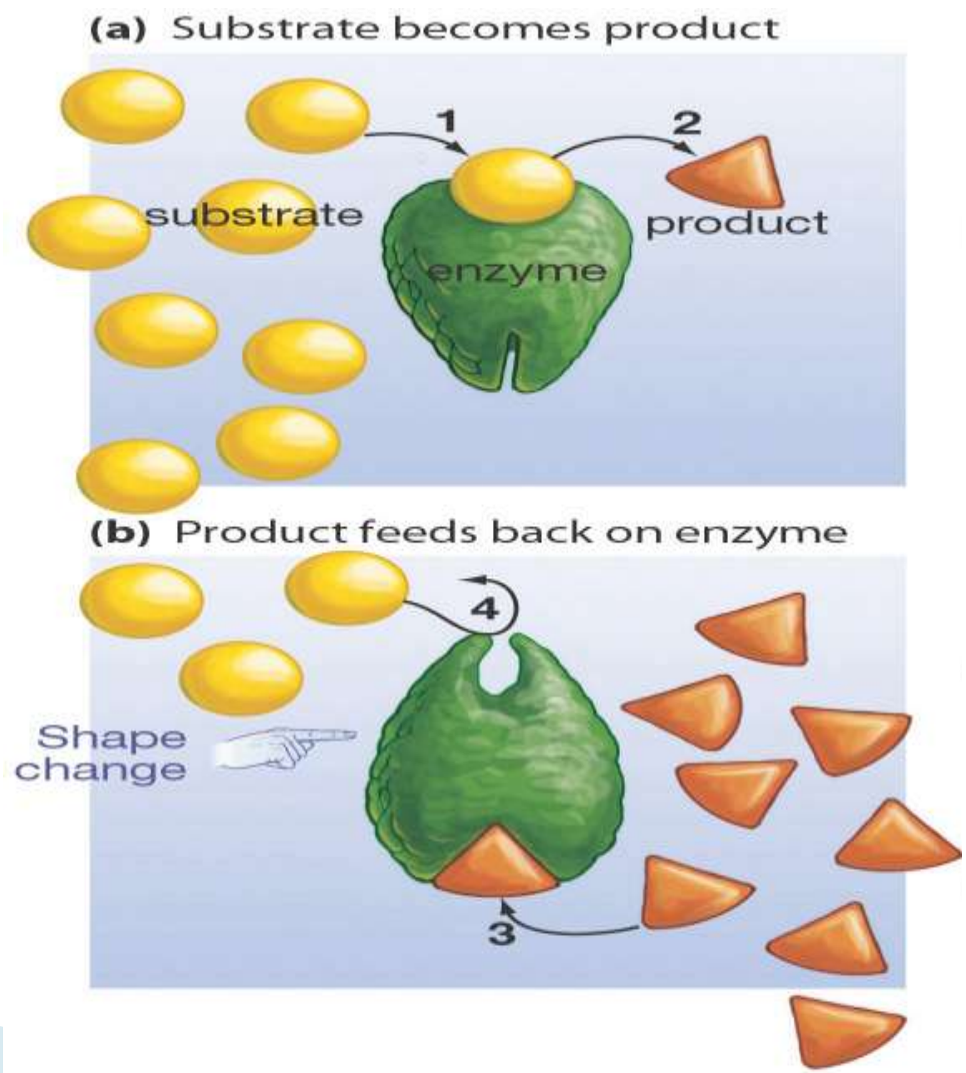
$$A \xrightarrow{e_A} B \xrightarrow{e_B} C \xrightarrow{e_C} D \xrightarrow{e_D} E \xrightarrow{e_F} F$$

Inhibition

The end products are controlling their own rate of production
There is no build up of intermediates (B,C, D and E).Usually such end product inhibition is effected allosterically

Feed back Inhibition

Accumulated product binds at a site other than the active site to bring about conformational changes, so as to inhibit the binding of the substrate. The rate of reaction declines.



Regulation of Enzyme activity

- Regulation of enzyme activity is needed to maintain homeostasis
- Many human diseases, including cancer, diabetes, cystic fibrosis, and Alzheimer's disease, are characterized by regulatory dysfunctions triggered by pathogenic agents or genetic mutations
- The flux of metabolites through metabolic pathways involves catalysis by numerous enzymes, active control of homeostasis is achieved by regulation of only a small number of enzymes
- Regulatory enzymes are usually the enzymes that are the rate-limiting, or committed step, in a pathway, meaning that after this step a particular reaction pathway will go to completion.

Regulation of Enzyme activity

- Two General Mechanisms that Affect Enzyme Activity:
- **1) control of the overall quantities of enzyme or concentration of substrates present**
- **2) alteration of the catalytic efficiency of the enzyme**

Regulation of Enzyme Concentrations

- The overall synthesis and degradation of a particular enzyme, also termed its **turnover number**, is one way of regulating the quantity of an enzyme.
- The amount of an enzyme in a cell can be increased by increasing its rate of synthesis, decreasing the rate of its degradation, or both.

Regulation of Enzyme Concentration: Induction

- **Induction** –an increase caused by an effector molecule
- This can manifest itself at the level of gene expression, RNA translation, and post-translational modifications
- The actions of many hormones and/or growth factors on cells leads to an increase in the expression and translation of "new" enzymes not present prior to the signal.
- Inducible enzymes of humans include tryptophan pyrrolase, aminotransferase, enzymes of the urea cycle, HMG-CoA reductase, Glucokinase and cytochrome P450.

Regulation of Enzyme Concentration- Repression

- An excess of a metabolite may curtail synthesis of its cognate enzyme via **repression.**
- Both induction and repression involve cis elements, specific DNA sequences located upstream of regulated genes, and trans-acting regulatory proteins.

Regulation of Enzyme Concentrations: Degradation

- The **degradation** of proteins constantly occurs in the cell
- Protein degradation by proteases is compartmentalized in the cell in the lysosome (which is generally non-specific), or in macromolecular complexes termed proteasomes
- Degradation by proteasomes is regulated by a complex pathway involving transfer of a 76 aa polypeptide, ubiquitin, to targeted proteins. Ubiquination of protein targets it for degradation by the proteasome.
- Proteolytic degradation is an irreversible mechanism
- dysfunctions of the ubiquitin-proteasome pathway contribute to the accumulation of aberrantly folded protein species characteristic of several neurodegenerative diseases

Regulation of Catalytic Activity

- **Zymogen Activation-** Certain proteins are synthesized and secreted as inactive precursor proteins known as **proproteins.** The proproteins of enzymes are termed **proenzymes** or **zymogens.**
- Selective proteolysis converts a proprotein by one or more successive proteolytic "clips" to a form that exhibits the characteristic activity of the mature protein, eg, its enzymatic activity.
- The digestive enzymes pepsin, trypsin, and chymotrypsin (proproteins = pepsinogen, trypsinogen, and chymotrypsinogen, respectively), several factors of the blood clotting and blood clot dissolution cascades, complement and Kinin system are examples of Zymogen activation.

Enzyme/Substrate Compartmentation

- Compartmentation ensures metabolic efficiency & simplifies regulation
- Segregation of metabolic processes into distinct subcellular locations like the cytosol or specialized organelles (nucleus, endoplasmic reticulum, Golgi apparatus, lysosomes, mitochondria, etc.) is another form of regulation

Enzyme Regulation by Compartmentation

Plasma membrane	Amino acid transport systems, Na^+-K^+ ATPase
Cytosol	Glycolysis, glycogenesis and glycogenolysis, hexose monophosphate pathway, fatty acid synthesis, purine and pyrimidine catabolism, aminoacyl-tRNA synthetases
Mitochondria	Tricarboxylic acid cycle, electron transport and oxidative phosphorylation, fatty acid oxidation, urea synthesis
Nucleus	DNA and RNA synthesis
Endoplasmic reticul (rough and smoot	Protein synthesis, steroid synthesis, glycosylation, detoxification
Lysosomes	Hydrolases
Golgi apparatus	Glycosyl transferases, glucose-5-phosphatase, formation of plasma membrane and secretory vesicles
Peroxisomes	Catalase, D-amino acid oxidase, urate oxidase

Allosteric Enzymes

- These enzymes function through reversible, non-covalent binding of a regulatory metabolite at a site other than the catalytic, active site.
- When bound, these metabolites do not participate in catalysis directly, but lead to conformational changes in one part of an enzyme that then affect the overall conformation of the active site (causing an increase or decrease in activity, hence these metabolites are termed allosteric activators or allosteric inhibitors
- Most of the Allosteric enzymes have multiple subunits

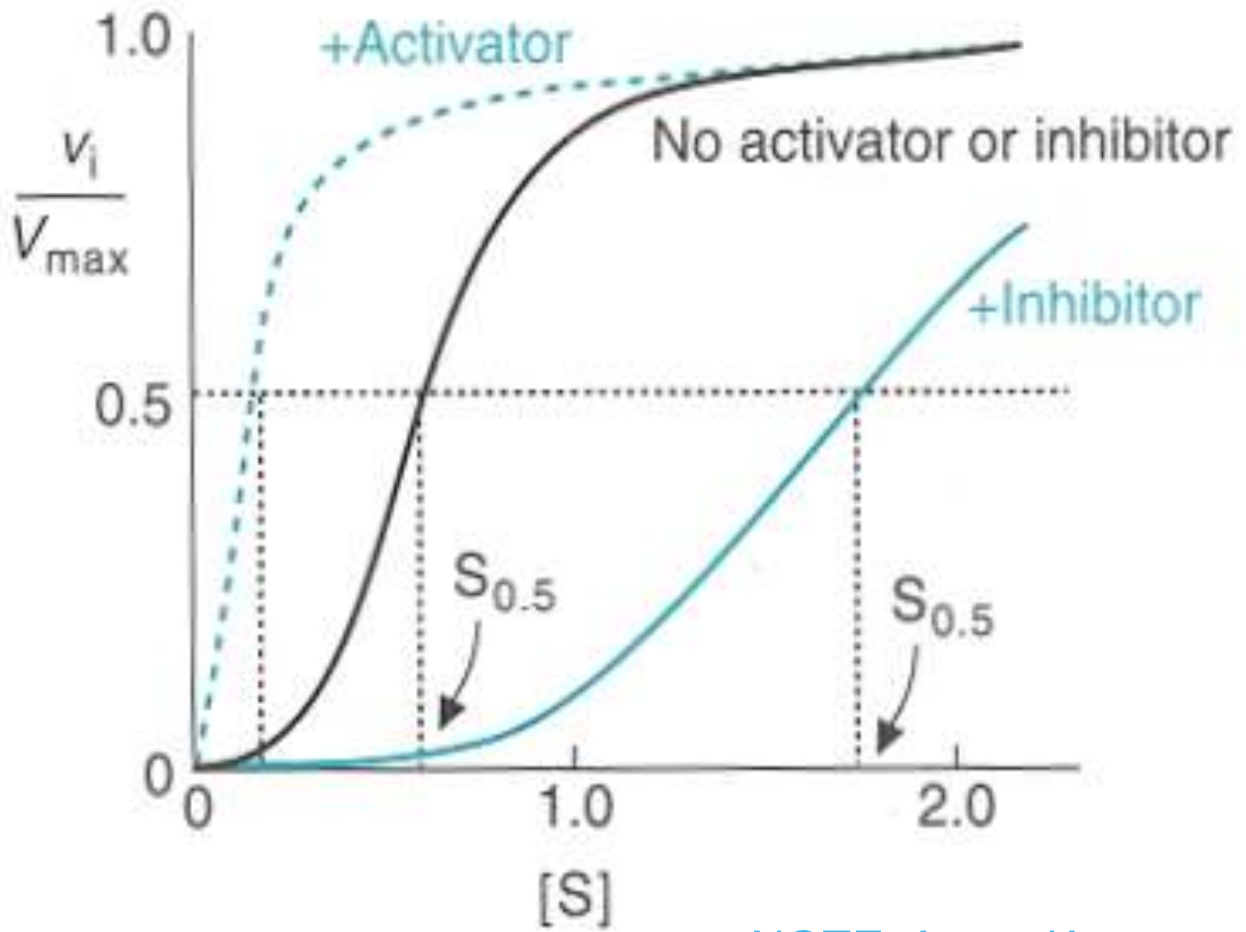
Allosteric Enzymes –Kinetics

- **Cooperativity** - in relation to multiple subunit enzymes, changes in the conformation of one subunit leads to conformational changes in adjacent subunits. These changes occur at the tertiary and quaternary levels of protein organization and can be caused by an allosteric regulator.
- **Homotropic regulation** - when binding of one molecule to a multi-subunit enzyme causes a conformational shift that affects the binding of the same molecule to another subunit of the enzyme.
- **Heterotropic regulation** - when binding of one molecule to a multi-subunit enzyme affects the binding of a different molecule to this enzyme (Note: These terms are similar to those used for oxygen binding to hemoglobin)

Allosteric Enzymes – Kinetics

- Allosteric enzymes do exhibit saturation kinetics at high [S], but they have a characteristic sigmoidal saturation curve rather than hyperbolic curve when v_o is plotted versus [S] (analogous to the oxygen saturation curves of myoglobin vs. hemoglobin). Addition of an allosteric activator (+) tends to shift the curve to a more hyperbolic profile (more like Michaelis-Menten curves), while an allosteric inhibitor (-) will result in more pronounced sigmoidal curves. The sigmoidicity is thought to result from the cooperativity of structural changes between enzyme subunits (again similar to oxygen binding to hemoglobin).

V_o vs [S] for Allosteric Enzymes



NOTE: A true Km cannot be determined for allosteric enzymes, so a comparative constant like $S_{0.5}$ or $K_{0.5}$ is used.

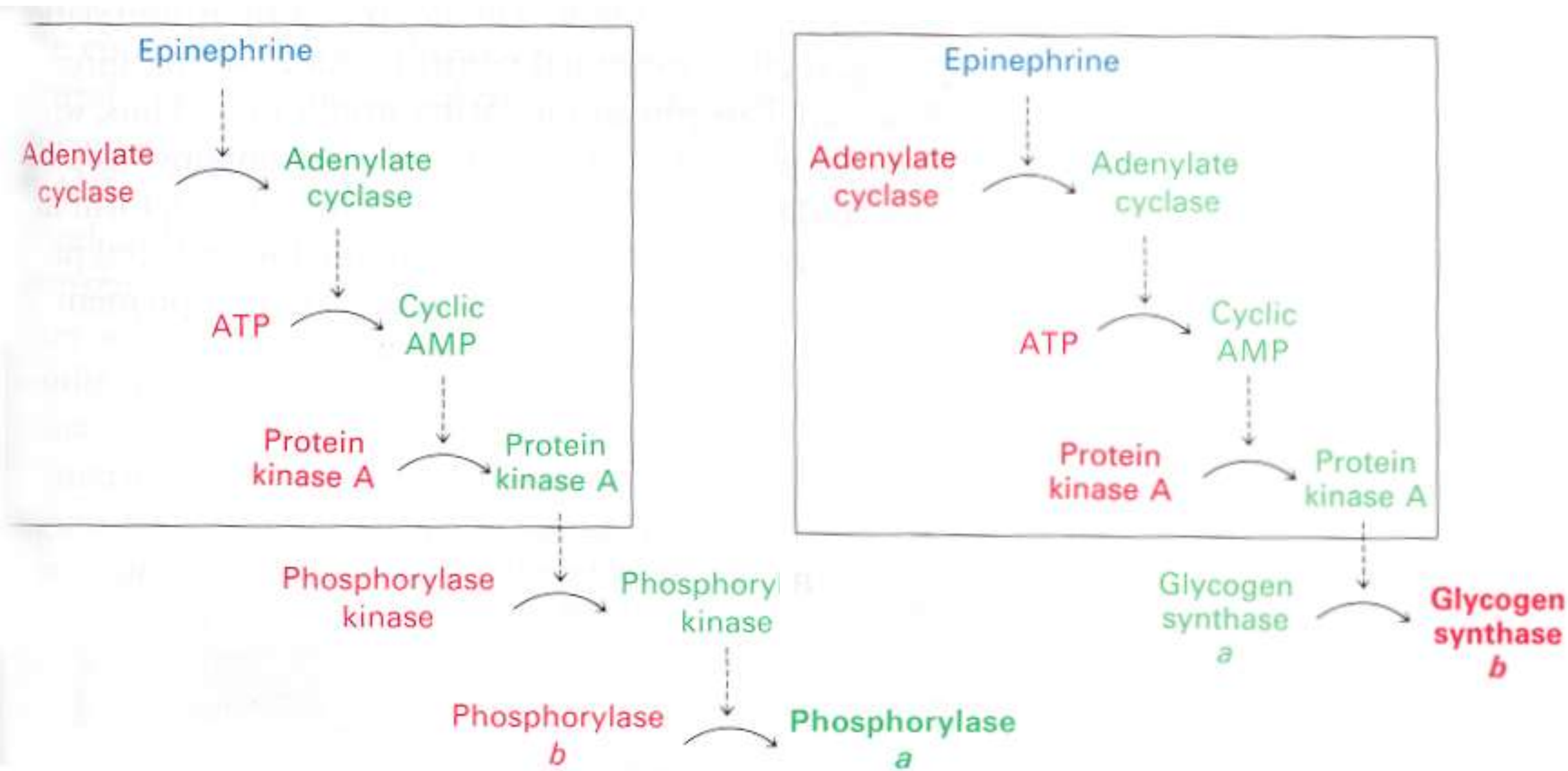
Regulation of Enzyme Activity by Covalent Modifications

- Another common regulatory mechanism is the reversible <u>covalent</u> modification of an enzyme. **Phosphorylation**, whereby a phosphate is transferred from an activated donor (usually ATP) to an amino acid on the regulatory enyme, is the most common example of this type of regulation. Frequently this phosphorylation occurs in response to some stimulus (like a hormone or growth factor) that will either activate or inactivate target enzymes

Phosphorylation/Signal Transduction

- Phosphorylation of one enzyme can lead to phosphorylation of a different enzyme which in turn acts on another enzyme, and so on. An example of this type of **phosphorylation cascade** is the response of a cell to cyclic AMP and its effect on glycogen metabolism.
- Use of a phosphorylation cascade allows a cell to respond to a signal at the cell surface and transmit the effects of that signal to intracellular enzymes (usually within the cytosol and nucleus) that modify a cellular process.
- This process is generically referred to as being part of a **signal transduction** mechanism

Signaling Regulation of Glycogen Synthase and Phosphorylase



A-forms, most active **B-forms, less active**

Feed back inhibition v/s feed back regulation

- Dietary cholesterol decreases hepatic synthesis of cholesterol, this feedback **regulation** does not involve feedback **inhibition.**
- HMG-CoA reductase, the rate-limiting enzyme of cholesterol genesis, is affected, but cholesterol does not feedback-inhibit its activity.
- Regulation in response to dietary cholesterol involves curtailment by cholesterol or a cholesterol metabolite of the expression of the gene that encodes HMG-CoA reductase (enzyme repression).